

INF-2204 Assignment1-report

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1 Introduction

In the last few years artificial intelligence(AI) has become more present within software development as its capabilities has improved. An often discussed issue with AI presence in software development is the reliability of the solutions it suggests. For this reason, the projects goal is to answer 3 key questions related to the before mentioned issue. 1.) Which model provides superior solutions, 2.) How reliable are the solutions when the complexity of the problem we are trying to solve increases and 3.) How well can AI be trusted overall when it comes to software development.

2 Related work

A paper by Russel Poldrack, Thomas Lu and Gasper Begus which covers AI-assisted coding with the GPT-4 model, that while it was a powerful tool, it had shortcomings in terms of validity and suggested that humans where still required to ensure the validity of the AI generated code. Considering the paper is written in 2023, this creates the question of whether this has changed as AI models have become more advanced or if their findings from 2023 still holds true.

3 System

The planned system to study are how well different AI models manages to provide good solutions in writing increasingly complex algorithms. Therefore, the system is limited to the code provided by the model. The idea for determining whether a solution is ideal or not, will be the overall correctness, if redundant code is present, and execution time if the provided solutions are different. For benchmarking the execution time, simple utilities will be implemented. Due to their popularity, two of the models for the experiment will be Claude and GPT. The idea is to also have a third model be part of the experiments, however, exactly which one is not yet decided. The choice will depend on popularity within coding (not necessarily overall popularity) as this will be more relevant to the purpose of the project.

4 Early environment and scope

Due to the projects nature, requirements in terms of hardware and software is fairly limited. Different hardware might be used for different experiments, however, all comparative benchmarks will run on the same hardware as to not give any benefits in terms of running on better hardware. Exactly which algorithms will be used for the experiments is not yet decided, meaning dataset requirements is currently unknown.

5 Early timeline

Initial work will consist of first deciding which AI models are going to be part of the experiment, and deciding exactly which algorithms should be used. As previously mentioned, execution time will be a relevant metric during experiments, therefore, a benchmarking framework will be implemented relatively early in the project. The exact design of the framework is not yet decided, as this might change depending on the needs of each experiment.

6 Discussion

When considering risks of choices, one in particular could prove problematic, token usage. Several models make use of a token system for requests. For non-paid solutions, the risk of running out of tokens when experimenting is a prevalent one. There are mitigating factors, swapping host sites/programs being the solution which will likely solve the issue altogether.

7 References

Russel A.Poldrack, Thomas Lu, Gasper Begus (2023). AI-assisted coding: Experiments with GPT-4